

1 Results

1.1 Logistic Regression Analysis of Mortality Predictors

A binary logistic regression was performed to ascertain the effects of age, ejection fraction, serum creatinine, and follow-up time on the likelihood of mortality in a clinical cohort ($N = 299$). The model was statistically significant, $\chi^2(4) = 119.27$, $p < .001$, and explained 42.4% (Nagelkerke R^2) of the variance in mortality events. The model correctly classified 82.1% of cases and demonstrated strong discriminative ability (ROC-AUC = 0.862).

Of the four predictors, three were statistically significant. Lower ejection fraction significantly increased the odds of mortality ($OR = 0.92$, 95% CI [0.89, 0.96], $p < .001$), as did elevated serum creatinine levels ($OR = 2.56$, 95% CI [1.60, 4.08], $p < .001$). Follow-up time exhibited a protective effect, with each additional day reducing the odds of mortality ($OR = 0.98$, 95% CI [0.97, 0.99], $p < .001$). Age was not a significant predictor in this multivariate model ($p = .085$). These findings suggest that hemodynamic and renal biomarkers are more immediate determinants of prognosis than chronological age in this specific cohort.